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On the Existence of Optimal $(v, 5, 1)$ and $(v, 6, 1)$ Binary Cyclically Permutable Constant-Weight Codes

Tsonka Baicheva [1,2,*] **and Svetlana Topalova** [1,2]

[1] Institute of Mathematics and Informatics, Bulgarian Academy of Sciences, 1113 Sofia, Bulgaria; svetlana@math.bas.bg
[2] Centre of Excellence in Informatics and Information and Communication Technologies, 1113 Sofia, Bulgaria
* Correspondence: tsonka@math.bas.bg

Abstract

The problem of the existence of optimal $(v, k, 1)$ binary cyclically permutable constant-weight (CPCW) codes has been completely solved for codeword weights $k < 5$. We consider the smallest open cases, namely $k = 5$ and $k = 6$. We present such codes for small values of the code length v and derive necessary conditions for the existence of optimal $(k(k-1)t + 2, k, 1)$ CPCW codes. These necessary conditions can be used to construct such codes, as well as to show that optimal codes with some parameters do not exist. In particular, we use them to prove that an optimal $(92, 6, 1)$ CPCW code does not exist.

Keywords: coding theory; binary cyclically permutable constant-weight code; optical orthogonal code; cyclic design

MSC: 94B60; 94B65

1. Introduction

1.1. Definitions

Denote by $\mathbb{Z}_v$ the additive group of integers modulo v. Let $C = \{c_0, c_1, \ldots, c_{k-1}\}$ be a k-element subset of $\mathbb{Z}_v$. Then, $\triangle C = \{c_i - c_j |\ i, j = 0, 1, \ldots, k-1; i \neq j\}$ is the multiset of differences of C, and $C + t$ denotes a t-translate of C, where $C + t = \{c_0 + t, c_1 + t, \ldots, c_{k-1} + t\}$ for $t \in \mathbb{Z}_v$.

Definition 1. *A $(v, k, 1)$ binary cyclically permutable constant-weight (CPCW) code of size s can be defined as a collection $\mathcal{C} = \{C_1, \ldots, C_s\}$ of k-subsets of $\mathbb{Z}_v$ (codewords) such that any two distinct translates of a codeword share at most one element, and any two translates of two distinct codewords also share at most one element:*

$$|C_i \cap (C_i + t)| \leq 1, \ \ 1 \leq i \leq s, \ \ 1 \leq t \leq v - 1 \tag{1}$$

$$|C_i \cap (C_j + t)| \leq 1, \ \ 1 \leq i < j \leq s, \ \ 0 \leq t \leq v - 1 \tag{2}$$

Everywhere in the text that follows, the abbreviation *CPCW* means "binary cyclically permutable constant weight". Originally, a CPCW code was defined as a collection of $\{0, 1\}$ sequences, but, for some purposes, it is easier to present the codewords as subsets of $\mathbb{Z}_v$, and that is why the above definition is widely used too. Condition (1) is called the auto-correlation property and (2) the cross-correlation property. We denote by s the *size* of $\mathcal{C}$, namely the number of its codewords.

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Consider a codeword $C = \{c_0, c_1, \ldots, c_{k-1}\}$. The auto-correlation property means that all the differences of a codeword of a $(v, k, 1)$ CPCW code are different, and the cross-correlation property means that $\triangle C_1 \bigcap \triangle C_2 = \emptyset$ for two codewords C_1 and C_2. Since $|\triangle C| = k(k-1)$, the size of a $(v, k, 1)$ CPCW code cannot exceed

$$\left\lfloor \frac{v-1}{k(k-1)} \right\rfloor.$$

CPCW codes that reach this bound are called *optimal* [1,2].

If the code size is exactly $(v-1)/k(k-1)$, the optimal $(v, k, 1)$ CPCW code is *perfect*; that is, all nonzero differences are covered.

CPCW codes are related to many other combinatorial structures. Definition 1, for instance, holds for a $(v, k, 1)$ optical orthogonal code (OOC) too [1]. In this paper, we will use the relation to partial designs.

Definition 2. *Let $V = \{P_i\}_{i=1}^{v}$ be a finite set of points and $\mathcal{B} = \{B_j\}_{j=1}^{b}$ be a finite collection of k-element subsets of V, called blocks. $D = (V, \mathcal{B})$ is a design (partial design) with parameters 2-(v,k,1) if any 2-subset of V is contained in exactly (at most) one block of $\mathcal{B}$. A 2-(v,k,1) design is also called a Steiner system and denoted by $S(2, k, v)$.*

Partial designs are also known as *packings* [3] or *packing designs* [4]. We call them partial designs following [5]. An *automorphism* of a 2-$(v, k, 1)$ partial design D is a permutation of its points that maps each block of D to a block of D.

Definition 3. *A 2-(v,k,1) design (partial design) is cyclic if it has an automorphism α permuting its points in one cycle, and it is strictly cyclic if each block orbit under this automorphism is of length v (no short orbits).*

A cyclic partial design with the maximum possible number of blocks is optimal.

Definition 4. *A circulant matrix (circulant) of order v is a (0,1) square matrix $M = (m_{i,j})_{v \times v}$ with v rows and columns such that $m_{i+1,j+1} = m_{i,j}$, $i, j \in \mathbb{Z}_v$.*

Perfect $(v, k, 1)$ CPCW codes are equivalent to strictly cyclic 2-$(v, k, 1)$ designs. An optimal $(v, k, 1)$ CPCW code is equivalent to an optimal 2-$(v, k, 1)$ strictly cyclic partial design. The incidence matrix of a 2-$(v, k, 1)$ strictly cyclic partial design contains submatrices, which are circulant matrices of order v. Each such circulant matrix corresponds to the translates of one of the codewords of the corresponding optimal $(v, k, 1)$ CPCW code. An example illustrating this equivalence is presented in Figure 1.

We are not going to define more related structures here. We shall only mention that optimal $(v, k, 1)$ CPCW codes correspond to $(v, k; \lfloor (v-1)/k(k-1) \rfloor)$ difference packings, and that perfect $(v, k, 1)$ CPCW codes are equivalent to $(v, k, 1)$ cyclic difference families. An interested reader can find more information, for instance, in [6,7].

1.2. Known Results and the Contribution of the Present Paper

Optimal $(v, k, 1)$ CPCW codes are studied in [2,8–10]. There are also many results on CPCW codes that were obtained for related combinatorial objects, like optical orthogonal codes [11–21], difference families [6,22–26], and combinatorial designs [27–30].

It was proved in [12] that an optimal $(v, 3, 1)$ CPCW code exists for all v except for $v = 6t + 2$ and $t \equiv 2$ or $3 \pmod 4$. Direct constructions with explicit codewords are presented in [21] to show the existence of an optimal (v,4,1) CPCW (OOC) code for any

positive integer $v \neq 25$. The existence of an optimal $(v, k, 1)$ CPCW code for $k \geq 5$ is still an open problem.

In this paper, we consider $k = 5$ and 6.

(**a**) Optimal (43,5,1) CPCW code

$\mathcal{C}$	$\triangle C$
{0, 1, 3, 7, 19}	{1, 2, 3, 4, 6, 7, 12, 16, 18, 19, 24, 25, 27, 31, 36, 37, 39, 40, 41, 42}
{0, 5, 13, 22, 33}	{5, 8, 9, 10, 11, 13, 15, 17, 20, 21, 22, 23, 26, 28, 30, 32, 33, 34, 35, 38}

Uncovered differences: 0, 14, 29

(**b**) Related optimal partial cyclic 2-(43,5,1) design

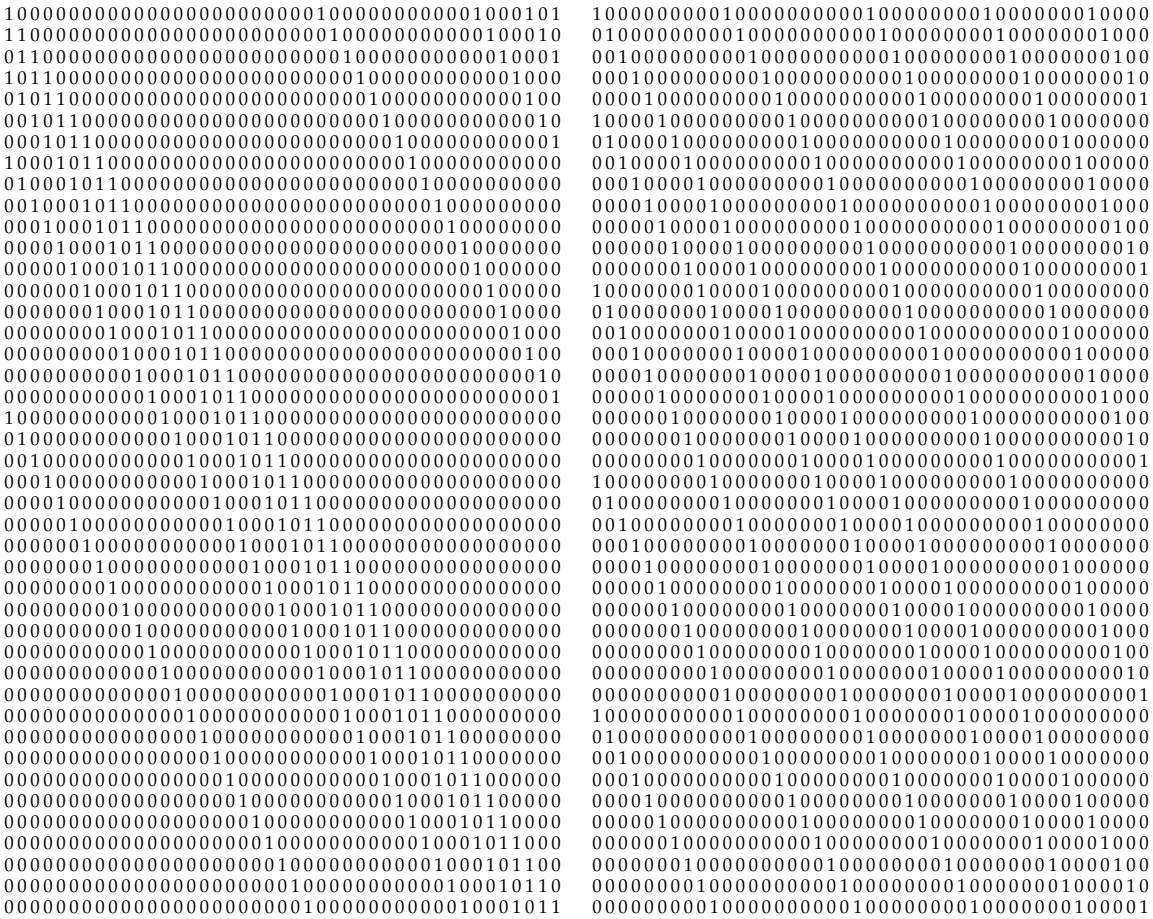


Figure 1. The corresponding partial design.

First, we will summarize the known results for optimal CPCW codes or related combinatorial objects for $k = 5$ and 6.

- An optimal $(15p, 5, 1)$ OOC with p a product of primes congruent to 1 modulo 4 and greater than 5 exists according to [18].
- It is shown in [14] that an optimal $(4p, 5, 1)$ OOC exists for prime $p \equiv 1 \pmod{10}$, and that there is an optimal $(4up, 5, 1)$ OOC for $u = 2, 3$ and prime $p \equiv 11 \pmod{20}$.
- A proof of the existence of a $(3^s 5v, 5, 1)$ OOC if v is a product of primes congruent to 1 modulo 4 and s is a non-negative integer is given in [16,17].
- There are also several direct constructions for optimal $(gv, 5, 1)$ OOCs where $60 \leq g \leq 180$, $g \equiv 0 \pmod{20}$, and v is a product of primes greater than 5.

- A class of optimal $(gv, 6, 1)$ optical orthogonal codes with v a product of primes congruent to 7 modulo 12 and greater than 7 and $g = 15, 20, 105, 140$ is obtained in [19].
- Some combinatorial constructions for optimal $(v, k, 1)$ OOCs using g-regular cyclic packings are derived in [4,20].

Some existence results about CPCW codes with $k = 5$ and 6 follow from results about cyclic difference families and combinatorial designs.

- It is shown in [23] that a $(p, 5, 1)$ difference family exists for $p \equiv 1 \pmod{20}$ for prime p.
- According to [26], a $(20t + 1, 5, 1)$ difference family exists for $1 \leq t \leq 50$ except possibly for $t = 16, 25, 31, 34, 40, 45$.
- The existence of a cyclic $(pq, 5, 1)$ difference family whenever p and q are primes congruent to $11 \pmod{20}$ is proved in [6].
- Existence of a $(q, 6, 1)$ cyclic difference family for any prime power $q \equiv 1 \pmod{30}$ with the exception of $q = 61$ is shown in [22].
- It was proved in [29] that a cyclic $(5p, 5, 1)$ combinatorial design exists for all $p \equiv 1 \pmod{4}$ and $p \neq 5$.

Nonexistence results for optimal $(v, 5, 1)$ OOCs with $v \equiv 22 \pmod{40}$ and for an optimal $(62, 6, 1)$ OOC are presented in [7].

Classification results about $(v, k, 1)$ CPCW codes for $k = 5$ and 6 or about related combinatorial objects can be found in [25,27,28,30,31]. The results obtained in these works show that, when v grows, the number of nonequivalent codes becomes extremely large. This makes the classifications for the next larger lengths practically unusable.

In this work, we consider $k = 5$ and 6 and the first values of v for which existence or nonexistence results are not known and try to find an example of an optimal $(v, k, 1)$ CPCW code for these parameters. We conduct this for $k = 5$ and $v \leq 181$ and $k = 6$ and $v \leq 151$. We cannot find an optimal code for 9 of the considered parameter sets. We notice that two of these gaps are for $v = k(k-1)t + 2$. This motivates us to consider the properties of such codes, and we derive necessary conditions for their existence and prove that an optimal $(92, 6, 1)$ CPCW code does not exist.

The paper is organized as follows: Section 2 demonstrates the basis of our approach. Section 3 presents both the computer-aided and analytical results that we obtain. Section 4 contains final remarks and open problems.

2. Methods

The explicit examples of new optimal $(v, 5, 1)$ and $(v, 6, 1)$ CPCW codes are obtained by the algorithm for classification of such codes, which we have used and described in [31]. It implies the construction of an array L with all possible codewords, namely with k-subsets of $\mathbb{Z}_v$ for which the auto-correlation property holds. We then construct codes by choosing their codewords from L in all possible ways. For that purpose, backtrack search on the elements of L is applied, and some partial solutions are filtered away if there exists an automorphism of $\mathbb{Z}_v$ that maps the partial solution to a partial solution that has already been constructed.

Our aim in the present work is not to obtain all the codes with these parameters but to find one such code. That is why we let the computation stop after having constructed one code, or after having worked for several days on a personal computer. The fact that we have not succeeded to construct a code with definite parameters does not mean that such a code does not exist.

The necessary conditions for the existence of optimal $(k(k-1)t + 2, k, 1)$ CPCW codes are derived using a method that is very popular in Design Theory, namely the counting of points and pairs of points in two different ways. The nonexistence of an optimal $(92, 6, 1)$

CPCW code is proved by using these necessary conditions and applying counting of the pairs of points in one more way.

3. Results

3.1. Examples of Optimal $(v,5,1)$ and $(v,6,1)$ CPCW Codes

We present in Tables 1 and 2 all the previously known existence and nonexistence results regarding optimal $(v,5,1)$ and $(v,6,1)$ CPCW codes with $v \leq 181$ and 151, respectively. Normal font is used for the v with theoretical results from [6–24,26,29] and italics for the results from the computer classifications in [25,27,28,30,31]. A normal or an italic v is struck through if the corresponding result (theoretic or computer-aided) shows nonexistence of optimal $(v,5,1)$ or $(v,6,1)$ CPCW codes. A question mark means that nothing is known about the existence of optimal codes with this length.

Table 1. Optimal $(v,5,1)$ CPCW codes before the present study.

41	~~*42*~~	*43*	~~44~~	45	*46–60*	61	~~62~~	*63–64*	65	*66–80*
81	*82–87*	88	89	?	101	~~102~~	?	105	?	121
?	124	125	?	132	?	135	?	141	~~142~~	?
145	?	161	?	164	165	?	181	~~*182*~~	?	185

Normal font for theoretical results, italic for computer-aided, strikethrough for nonexistence and symbol ? when nothing is known.

Table 2. Optimal $(v,6,1)$ CPCW codes before the present study.

~~*61*~~	~~62~~	*63*	~~*64*~~	~~*65*~~	*66–91*	~~*92*~~	*93–105*	?	121	?	151

Normal font for theoretical results, italic for computer-aided, strikethrough for nonexistence and symbol ? when nothing is known.

Tables 3 and 4 are organized in a similar way to Tables 1 and 2, but in them the existence results from the present paper are implemented too, and bold font is used for them. The question marks mean that we could not find optimal codes for these values of v and the question of the existence of such codes remains open. In particular, we have not found examples of optimal $(v,5,1)$ CPCW codes for $v = 143, 144, 146, 162$, and 167, and of optimal $(v,6,1)$ CPCW codes for $v = 122, 123, 124$, and 126. These are $(k(k-1)t+c,k,1)$ CPCW codes with relatively small c. The latter is also small for the optimal codes for which there are no theoretical nonexistence results, but they have been proved not to exist with the help of a computer [31], namely an optimal $(42,5,1)$ CPCW code and optimal $(v,6,1)$ CPCW codes for $v = 61, 64, 65$, and 92. This motivated our study of the properties of optimal $(k(k-1)t+2,k,1)$ CPCW codes.

Table 3. New optimal $(v,5,1)$ CPCW codes.

41	~~*42*~~	*42–43*	44	45	*46–60*	61	~~62~~	*63–64*	65	*66–80*
81	*82–87*	88	89	**90–100**	101	~~102~~	**103–104**	105	**106–120**	121
122–123	124	125	**126–131**	132	**133–134**	135	**136–140**	141	~~142~~	?
145	?	**147–160**	161	?	164	165	**166**	?	**168–180**	181

Bold font for results in the present paper, normal font for theoretical results, italic for computer-aided, strikethrough for nonexistence and symbol ? when nothing is known.

Table 4. New optimal $(v, 6, 1)$ CPCW codes.

~~61~~	~~62~~	*63*	~~64~~	~~65~~	*66–91*	~~92~~	*93–105*	**106–120**	121	?	**125**	?	**127–150**	151

Bold font for results in the present paper, normal font for theoretical results, italic for computer-aided, strikethrough for nonexistence and symbol ? when nothing is known.

Tables A1 and A2 in Appendix A contain all the examples of new optimal $(v, 5, 1)$ and $(v, 6, 1)$ CPCW codes. Each codeword contains the element $0 \in \mathbb{Z}_v$, and therefore this element is not presented.

3.2. Properties of Optimal $(k(k-1)t+2, k, 1)$ CPCW Codes

The optimal $(v, k, 1)$ CPCW codes of length $k(k-1)t+1$ correspond to 2-$(k(k-1)t+1, k, 1)$ designs. They are perfect because their codewords cover all $v-1$ differences. Optimal CPCW codes of length $k(k-1)t+2$, however, have remarkable combinatorial properties too because they do not cover only one difference. We derive here some necessary conditions for the existence of such codes.

Lemma 1. *The element $\frac{v}{2} \in \mathbb{Z}_v$ cannot be covered by the differences of the codewords of a $(v, k, 1)$ CPCW code of even length. For even v, the element $\frac{v}{2} \in \mathbb{Z}_v$ can be obtained as the difference $a-b$ of $\frac{v}{2}$ different pairs $\{a, b\}$.*

Proof. Let C be a codeword of a $(v, k, 1)$ CPCW code $\mathcal{C}$ of even length. Let $a, b \in C$. Then, $a-b$ and $b-a$ must be two different differences from $\triangle C$. Suppose $a-b=\frac{v}{2}$, and then $b-a=v-\frac{v}{2}=\frac{v}{2}$, and $\frac{v}{2}$ appears twice in $\triangle C$, but this is impossible for a $(v, k, 1)$ CPCW code. Since $a-b=b-a=\frac{v}{2}$, the element $\frac{v}{2} \in \mathbb{Z}_v$ can be obtained as the difference $a-b$ of the $\frac{v}{2}$ different pairs $\{a, a+\frac{v}{2}\}$, $a \leq \frac{v}{2}$. □

Lemma 2. *The codewords of an optimal $(k(k-1)t+2, k, 1)$ CPCW code leave uncovered only the difference $\frac{v}{2}$.*

Proof. Since $k(k-1)t+2$ is even, by Lemma 1, the possible differences are $v-2=k(k-1)t$. The size of such a code is

$$\left\lfloor \frac{v-1}{k(k-1)} \right\rfloor = \left\lfloor \frac{k(k-1)t+1}{k(k-1)} \right\rfloor = t,$$

and each codeword covers $k(k-1)$ differences. Then, all the codewords cover $k(k-1)t=v-2$ differences; that is, only one difference remains uncovered. □

An optimal $(k(k-1)t+2, k, 1)$ CPCW code corresponds to an optimal partial cyclic 2-$(k(k-1)t+2, k, 1)$ design. The translates of each codeword correspond to v blocks of this partial design and to a circulant submatrix of the incidence matrix of the partial design. We next consider some properties of this partial design.

Lemma 3. *The $\frac{v}{2}$ pairs of points $\{a, a+\frac{v}{2}\}$ are not contained in the blocks of an optimal partial cyclic 2-$(k(k-1)t+2, k, 1)$ design corresponding to an optimal $(k(k-1)t+2, k, 1)$ CPCW code. Each of the remaining $\frac{v(v-2)}{2}$ pairs of points is in exactly one block of this partial design.*

Proof. The $\frac{v}{2}$ pairs of points that are not contained in the blocks of the design correspond to the $\frac{v}{2}$ different pairs $\{a, a+\frac{v}{2}\}$ from Lemma 1. □

Lemma 4. *Denote the rows/columns of a circulant M of an even order v by the numbers $0, 1, 2, \ldots v-1$. Let φ be a permutation that rearranges the rows and the columns of M in*

such a way that they follow each other as $0, 2, 4 \ldots v-2, 1, 3, 5, \ldots v-1$. *Then, the resulting matrix* φM *is composed of* 4 *circulants* A, B, C, *and* D *of order* $\frac{v}{2}$.

$$\varphi M = \begin{pmatrix} A & B \\ C & D \end{pmatrix}.$$

Denote by u_A, u_B, u_C, *and* u_D *the number of ones in a row/column of the matrices* A, B, C, *and* D, *respectively. Then,* $u_A = u_D$ *and* $u_B = u_C$.

Proof. Cutting off the even rows and even columns of M leads to a matrix A of dimension $\frac{v}{2} \times \frac{v}{2}$. Let $m_{i,j}$ be the element in row i and column j of M and $a_{i,j}$ be the element in row i and column j of A. Then, $a_{i,j} = m_{2i,2j}$.

We want to show that A is a circulant matrix. Addition of the indices of A is modulo $\frac{v}{2}$, and of M modulo v. Since $2i \pmod{v} = 2(i \pmod{\frac{v}{2}})$, the addition of the even indices of M can be considered modulo $\frac{v}{2}$ too. Then, $a_{i+1,j+1} = m_{2i+2,2j+2}$. M is a circulant, and therefore $m_{2i+2,2j+2} = m_{2i,2j} = a_{i,j}$. It follows that $a_{i+1,j+1} = a_{i,j}$; that is, A is a circulant matrix.

Similarly, the circulants B, C, and D are obtained by cutting off the even rows and odd columns, the odd rows and even columns, and the odd rows and odd columns, respectively.

The number of ones in each row and column of φM is the same, and therefore $u_A = u_D$ and $u_B = u_C$. □

Example 1. *We illustrate the action of* φ *on two arbitrary circulants of even orders* 10 *and* 8.

	0 1 2 3 4 5 6 7 8 9
0	0 1 0 0 0 1 0 1 0 0
1	0 0 1 0 0 0 1 0 1 0
2	0 0 0 1 0 0 0 1 0 1
3	1 0 0 0 1 0 0 0 1 0
4	0 1 0 0 0 1 0 0 0 1
5	1 0 1 0 0 0 1 0 0 0
6	0 1 0 1 0 0 0 1 0 0
7	0 0 1 0 1 0 0 0 1 0
8	0 0 0 1 0 1 0 0 0 1
9	1 0 0 0 1 0 1 0 0 0

	0 1 2 3 4 5 6 7 8 9
0	0 1 0 0 0 1 0 1 0 0
2	0 0 0 1 0 0 0 1 0 1
4	0 1 0 0 0 1 0 0 0 1
6	0 1 0 1 0 0 0 1 0 0
8	0 0 0 1 0 1 0 0 0 1
1	0 0 1 0 0 0 1 0 1 0
3	1 0 0 0 1 0 0 0 1 0
5	1 0 1 0 0 0 1 0 0 0
7	0 0 1 0 1 0 0 0 1 0
9	1 0 0 0 1 0 1 0 0 0

	0 2 4 6 8	1 3 5 7 9
0	0 0 0 0 0	1 0 1 1 0
2	0 0 0 0 0	0 1 0 1 1
4	0 0 0 0 0	1 0 1 0 1
6	0 0 0 0 0	1 1 0 1 0
8	0 0 0 0 0	0 1 1 0 1
1	0 1 0 1 1	0 0 0 0 0
3	1 0 1 0 1	0 0 0 0 0
5	1 1 0 1 0	0 0 0 0 0
7	0 1 1 0 1	0 0 0 0 0
9	1 0 1 1 0	0 0 0 0 0

	0 1 2 3 4 5 6 7
0	0 1 0 1 0 0 1 0
1	0 0 1 0 1 0 0 1
2	1 0 0 1 0 1 0 0
3	0 1 0 0 1 0 1 0
4	0 0 1 0 0 1 0 1
5	1 0 0 1 0 0 1 0
6	0 1 0 0 1 0 0 1
7	1 0 1 0 0 1 0 0

	0 1 2 3 4 5 6 7
0	0 1 0 1 0 0 1 0
2	1 0 0 1 0 1 0 0
4	0 0 1 0 0 1 0 1
6	0 1 0 0 1 0 0 1
1	0 0 1 0 1 0 0 1
3	0 1 0 0 1 0 1 0
5	1 0 0 1 0 0 1 0
7	1 0 1 0 0 1 0 0

	0 2 4 6	1 3 5 7
0	0 0 0 1	1 1 0 0
2	1 0 0 0	0 1 1 0
4	0 1 0 0	0 0 1 1
6	0 0 1 0	1 0 0 1
1	0 1 1 0	0 0 0 1
3	0 0 1 1	1 0 0 0
5	1 0 0 1	0 1 0 0
7	1 1 0 0	0 0 1 0

Lemma 5. *Let* M *be the incidence matrix of an optimal partial cyclic* 2-$(k(k-1)t+2, k, 1)$ *design* D *corresponding to an optimal* $(k(k-1)t+2, k, 1)$ *CPCW code. If the permutation* φ *from Lemma 4 is applied to the rows of* M *and to the columns of each of its* t *circulant submatrices, a matrix with* $4t$ *circulant submatrices of order* $\frac{v}{2}$ *is obtained. If* $v \equiv 2 \pmod{4}$, *the columns of each of the two rows of submatrices cover all possible* $\frac{v(v-2)}{8}$ *pairs of points, and the missing* $\frac{v}{2}$ *pairs of points in the partial design have one point from the first and one from the second row of*

submatrices. If $v \equiv 0 \pmod 4$, *there are* $\frac{v}{4}$ *pairs of points that are not covered by the columns of each row of submatrices, and all possible* $\frac{v^2}{4}$ *pairs of points having one point from the first and one from the second row of submatrices are covered.*

Proof. By Lemma 4, the permutation φ transforms each of the t circulants of order v to 4 circulants of order $\frac{v}{2}$. That is why the incidence matrix φM of the partial design has $4t$ circulant submatrices of order $\frac{v}{2}$. If $v \equiv 2 \pmod 4$, then $\frac{v}{2}$ is odd. That is why, if a is odd, $a + \frac{v}{2}$ is even and vice versa. Since the first row of submatrices of φM contains the even rows of M and the second the odd rows, the missing pairs of points $\{a, a + \frac{v}{2}\}$ (Lemma 3) have one point from the first row of submatrices and one from the second. If $v \equiv 0 \pmod 4$, then $\frac{v}{2}$ is even, a and $a + \frac{v}{2}$ are both even or both odd, and the missing pairs of points consist of two points from one and the same row of submatrices. □

With respect to the number of ones in the four submatrices, φ can transform a circulant of order v to $\left\lfloor \frac{k}{2} \right\rfloor + 1$ types of matrices $A_0, A_1, \ldots A_{\left\lfloor \frac{k}{2} \right\rfloor}$ such that A_i has two circulant submatrices with i ones in each row/column and two with $k - i$ ones in each row/column.

Example 2. *Types of matrices for* $k = 5$ *and* 6

For $k = 5$, *the types of matrices are*

A_0	A_1	A_2
0 5 *5 0*	*1 4* *4 1*	*2 3* *3 2*

For $k = 6$, *the types of matrices are*

A_0	A_1	A_2	A_3
0 6 *6 0*	*1 5* *5 1*	*2 4* *4 2*	*3 3* *3 3*

Theorem 1. *For an optimal* $(v, k, 1)$ *CPCW code with* $v = k(k-1)t + 2$ *and* $v \equiv 2 \pmod 4$, *the following equations hold:*

$$\sum_{i=0}^{\frac{v}{2}} a_i = \frac{v-2}{k(k-1)} \tag{3}$$

$$\sum_{i=0}^{\frac{v}{2}} \left(\frac{i(i-1)}{2} + \frac{(k-i)(k-i-1)}{2} \right) a_i = \frac{v-2}{4} \tag{4}$$

$$\sum_{i=1}^{\frac{v}{2}} i(k-i)a_i = \frac{v-2}{4} \tag{5}$$

Proof. We obtain the equations by counting in different ways the pairs of points in φM. Denote by a_i the number of $v \times v$ matrices of type A_i that are contained in φM. Their number is $t = \frac{v-2}{k(k-1)}$, and this is Equation (3). Equation (4) counts the pairs of points from one and the same row of submatrices. By Lemma 5, their number is $\frac{v(v-2)}{8}$. On the other hand, each $\frac{v}{2} \times \frac{v}{2}$ submatrix of a matrix of type A_i covers $\frac{v}{2}\left(\frac{i(i-1)}{2} + \frac{(k-i)(k-i-1)}{2}\right)$ pairs of points. Equation (5) counts the pairs in which the two points are from different rows of submatrices. By Lemma 5, all but the missing $\frac{v}{2}$ pairs are covered; that is, all such pairs are $\frac{v^2}{4} - \frac{v}{2} = \frac{v(v-2)}{4}$. On the other hand, each $v \times v$ submatrix of type A_i covers $i(k-i)v$ such pairs of points. □

Theorem 2. *For an optimal* $(v, k, 1)$ *CPCW code with* $v = k(k-1)t + 2$ *and* $v \equiv 0 \pmod 4$, *the following equations hold:*

$$\sum_{i=0}^{\frac{v}{2}} a_i = \frac{v-2}{k(k-1)} \tag{6}$$

$$\sum_{i=0}^{\frac{v}{2}} \left(\frac{i(i-1)}{2} + \frac{(k-i)(k-i-1)}{2} \right) a_i = \frac{v-4}{4} \tag{7}$$

$$\sum_{i=1}^{\frac{v}{2}} i(k-i)a_i = \frac{v}{4} \tag{8}$$

Proof. The proof is similar to that of Theorem 1, but, by Lemma 5, the pairs of points from one and the same row of submatrices are $\frac{v(v-2)}{8} - \frac{v}{4} = \frac{v(v-4)}{8}$. All the pairs in which the two points are from different rows of submatrices are covered. That is why their number is $\frac{v^2}{4}$. □

Corollary 1. *An optimal* $(20t + 2, 5, 1)$ *CPCW code can only exist if* t *is even.*

Proof. In this case, $v \equiv 2 \pmod 4$, and Equation (5) gives

$$4a_1 + 6a_2 = \frac{v-2}{4}.$$

That is why $v - 2$ should be divisible by 8. □

Corollary 1 is not a new result. It is part of the nonexistence results presented in [7], Theorem 4.6. The next example shows the application of Theorem 1 on an optimal $(20t + 2, 5, 1)$ CPCW code with even t.

Example 3. *For an optimal* $(82, 5, 1)$ *CPCW code by Theorem 1:*

$$a_0 + a_1 + a_2 + a_3 = 4 \tag{9}$$

$$5a_0 + 3a_1 + 2a_2 = 10 \tag{10}$$

$$2a_1 + 3a_2 = 10 \tag{11}$$

There exists a unique integer solution: $a_0 = 0, a_1 = 2, a_2 = 2$, *and the optimal* $(82, 5, 1)$ *CPCW codes comply with it.*

3.3. Nonexistence of an Optimal $(92, 6, 1)$ *CPCW Code*

Multiple nonexistence results on optimal $(v, k, 1)$ CPCW codes are obtained in [7]. They include the nonexistence of an optimal $(62, 6, 1)$ code. We do not know other theoretical nonexistence results for optimal $(30t + 2, 6, 1)$ CPCW codes. Our computer-aided classification in [31] shows that an optimal $(92, 6, 1)$ CPCW code does not exist. Using Theorem 2, we shall now prove this without a computer.

Proposition 1. *An optimal* $(92, 6, 1)$ *CPCW code does not exist.*

Proof. By Theorem 2:

$$a_0 + a_1 + a_2 + a_3 = 3 \tag{12}$$

$$15a_0 + 10a_1 + 7a_2 + 6a_3 = 22 \tag{13}$$

$$5a_1 + 8a_2 + 9a_3 = 23 \tag{14}$$

There exists a unique integer solution:

$$a_0 = 0, a_1 = 1, a_2 = 0, a_3 = 2 \tag{15}$$

It follows that the number of ones in φM is distributed as

$$\begin{array}{ccc} 5\,1 & 3\,3 & 3\,3 \\ 1\,5 & 3\,3 & 3\,3 \end{array}$$

We can see from (15) that the submatrices of dimension 46×46 have $1, 3$, or 5 ones in each row/column. Since 46 is even and $v/4 = 23$ is odd, we can use the same ideology as in Lemma 5 and Theorem 1 on the variable-length code corresponding to the first 46 rows of φM; that is, we apply to the first 46 rows of φM a permutation φ' such that after its application these rows follow each other as $0, 2, 4 \ldots 44, 1, 3, 5, \ldots 45$, and we apply the same permutation to the columns of the submatrices of dimension 46.

A 46×46 circulant with 1, 3, or 5 ones in each row/column can be transformed by φ' to one, two, or three types of matrices, respectively. We denote them $A_{1,0}$,$A_{3,0}$, $A_{3,1}$, $A_{5,0}$, $A_{5,1}$, and $A_{5,2}$. The matrix $A_{i,j}$ is of dimension 46×46 and has four submatrices of dimension 23×23, two of them with j and two with $i - j$ ones in each row/column.

$$\begin{array}{cccccc} A_{1,0} & A_{3,0} & A_{3,1} & A_{5,0} & A_{5,1} & A_{5,2} \\ 0\,1 & 0\,3 & 1\,2 & 0\,5 & 1\,4 & 2\,3 \\ 1\,0 & 3\,0 & 2\,1 & 5\,0 & 4\,1 & 3\,2 \end{array}$$

Denote by $a_{i,j}$ the number of 23×23 matrices of type $A_{i,j}$ that are contained in the first 23 rows of $\varphi' \varphi M$. Since 23 is odd, we can apply Theorem 1 and write an analogue of Equation (5) for $a_{i,j}$; thus, we obtain

$$\sum_{(i,j) \in S} j(i-j)a_{i,j} = \frac{46-2}{4} = 11 \tag{16}$$

where $S = \{(1,0), (3,0), (3,1), (5,0), (5,1), (5,2)\}$. It follows that

$$2a_{3,1} + 4a_{5,1} + 6a_{5,2} = 11. \tag{17}$$

The same result can be obtained by counting the pairs of points such that one is from the first 23 rows of $\varphi' \varphi M$ and the other from the second 23 rows. The missing 23 pairs of points are among them, and therefore

$$46(2a_{3,1} + 4a_{5,1} + 6a_{5,2}) = 23 \cdot 23 - 23 = 23 \cdot 22 = 46 \cdot 11.$$

This results in Equation (17) again. Equation (17), however, has no integer solutions because its left side is even and the right one is odd. That is why an optimal $(92, 6, 1)$ CPCW code cannot exist. □

4. Discussion

The presented examples of optimal $(v, 5, 1)$ and $(v, 6, 1)$ CPCW codes might be useful in relevant applications, as well as in future research on the topic. The results for small parameters show that nonexistence of some optimal $(k(k-1)t + c, k, 1)$ CPCW codes with small c can be expected. The solutions of the equations from Theorems 1 and 2 might be useful in further theoretical or computer-aided research on optimal $(k(k-1)t + 2, k, 1)$ CPCW codes whose existence remains undecided.

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Abbreviations

The following abbreviations are used in this manuscript:

CPCW Code	Binary Cyclically Permutable Constant-Weight Code
OOC	Optical Orthogonal Code
CDF	Cyclic Difference Family

Appendix A

Table A1. Optimal (v,5,1) CPCW codes.

Length	Codewords				
90	1 3 7 18	5 25 33 52	13 39 53 69	9 31 55 67	
91	1 3 7 12	8 33 49 73	14 35 52 71	10 23 54 69	
92	1 3 7 12	15 35 54 76	13 30 55 78	8 26 36 60	
93	1 3 7 12	8 27 40 73	10 26 48 78	14 31 49 70	
94	1 3 7 12	8 34 50 77	14 33 53 71	10 31 46 59	
95	1 3 7 15	9 32 53 84	13 41 59 78	5 29 39 55	
96	1 3 7 12	10 36 63 76	8 25 57 75	14 37 52 68	
97	1 3 7 12	10 30 53 70	13 32 47 68	8 24 46 72	
98	1 3 7 12	16 38 59 78	8 26 61 74	10 27 41 56	
99	1 3 7 12	10 34 51 72	14 29 57 83	8 33 55 68	
100	1 3 7 12	14 38 66 81	13 36 54 71	8 30 40 61	
103	1 3 15 96	9 49 65 92	4 37 50 79	6 41 58 77	5 30 64 85
104	1 3 7 90	9 27 63 82	10 35 47 81	5 45 65 93	8 38 51 80
106	1 3 8 43	11 48 60 79	13 34 67 89	9 45 59 74	4 20 82 100
107	1 3 7 34	9 23 51 71	10 32 50 96	8 24 37 63	5 35 54 95
108	1 3 7 39	9 26 56 93	10 31 53 95	8 19 33 68	5 34 50 62
109	1 3 7 31	8 22 33 65	9 35 48 58	5 50 68 97	15 53 69 90
110	1 3 7 27	8 42 72 101	12 25 40 73	5 36 52 96	10 21 53 75
111	1 3 7 21	12 41 65 84	9 44 78 89	10 47 62 98	8 25 63 79
112	1 3 7 16	10 45 62 88	8 19 49 74	12 44 73 91	5 28 42 64

Table A1. *Cont.*

Length	Codewords					
113	1 3 7 12	13 34 58 84	10 30 47 70	15 31 59 77	8 27 49 88	
114	1 3 7 15	9 41 59 86	10 21 52 81	17 51 75 95	5 40 66 89	
115	1 3 7 25	8 34 45 102	5 28 84 101	9 41 61 71	12 51 67 100	
116	1 3 7 17	8 30 90 101	5 36 54 75	12 59 78 91	9 29 64 92	
117	1 3 7 18	5 38 66 74	10 30 70 83	16 42 71 92	9 31 68 103	
118	1 3 7 24	9 27 63 98	11 33 61 77	12 31 79 93	5 15 45 58	
119	1 3 7 16	10 30 41 70	5 43 71 105	8 26 72 95	12 33 77 94	
120	1 3 7 24	13 39 72 91	9 46 66 102	10 41 53 85	8 58 73 98	
122	1 3 21 33	10 34 93 106	8 36 76 111	6 51 58 113	4 42 85 99	5 49 74 105
123	1 3 22 53	8 67 93 103	4 17 33 65	9 55 98 109	5 12 49 88	6 24 51 66
126	1 3 25 122	6 23 85 100	8 38 75 91	11 44 72 92	12 68 86 99	9 66 80 116
127	1 3 21 54	4 12 84 89	9 25 57 91	11 41 63 112	6 29 68 87	7 17 31 44
128	1 3 82 119	4 99 105 110	7 21 69 97	13 39 55 96	8 51 75 111	15 35 65 109
129	1 3 15 105	13 36 82 100	8 99 109 120	6 32 63 85	4 37 71 78	5 54 73 89
130	1 3 8 75	6 17 51 105	12 35 64 83	9 27 37 117	4 30 73 114	15 39 77 109
131	1 3 9 55	4 64 105 115	11 33 81 99	5 12 36 108	15 49 68 106	13 58 75 102
133	1 3 8 113	9 27 72 86	12 36 62 96	10 51 68 103	6 39 85 117	4 29 42 118
134	1 3 7 37	12 55 68 94	9 47 92 111	22 57 81 106	5 15 46 63	8 29 62 73
136	1 3 7 23	8 73 82 100	14 57 89 108	11 60 91 106	12 38 78 115	5 34 69 86
137	1 3 7 16	12 36 55 84	11 33 50 77	5 42 63 117	10 40 86 109	8 49 67 81
138	1 3 7 27	10 22 66 127	9 48 62 107	16 84 103 120	8 49 81 123	5 30 43 80
139	1 3 7 28	8 24 56 85	10 33 53 102	12 31 46 76	13 55 73 113	14 65 82 117
140	1 3 7 34	9 26 78 127	8 29 43 66	10 55 83 102	5 64 89 129	12 36 80 122
147	1 3 9 88 12 40 76 126	5 48 72 113	4 19 100 122	17 37 91 129	7 23 49 101	13 27 58 90
148	1 3 44 58 9 19 37 79	6 108 119 141	4 27 81 86	15 45 83 132	8 34 84 131	12 32 85 124
149	1 3 18 127 13 27 56 80	4 12 61 72	10 26 58 105	7 52 94 115	5 33 64 83	9 39 74 112
150	1 3 36 110 4 48 64 132	5 39 58 138	9 81 104 130	6 56 63 71	11 32 123 137	10 52 83 113
151	1 3 10 56 11 25 92 124	4 12 40 73	15 39 87 104	16 76 110 130	5 88 107 138	22 45 80 122
152	1 3 10 95 14 29 84 134	11 33 110 133	8 54 91 125	6 31 72 93	5 101 114 140	4 28 44 107
153	1 3 11 32 4 17 40 67	14 44 70 118	7 19 62 84	5 33 94 133	9 24 66 82	6 47 81 99

Table A1. *Cont.*

Length	Codewords					
154	1 3 10 100 4 16 30 82	11 50 71 129	8 49 89 135	6 35 67 126	5 20 53 137	18 56 80 103
155	1 3 7 62 22 47 82 127	12 36 84 124	13 39 57 91	8 27 38 114	17 46 102 135	5 21 63 86
156	1 3 7 87 17 41 94 130	5 15 35 123	9 55 73 105	8 29 57 71	16 39 61 98	19 66 100 131
157	1 3 7 15 19 42 75 99	9 27 63 135	16 48 83 112	17 51 98 119	11 52 118 131	10 30 70 114
158	1 3 7 33 11 67 80 129	9 27 63 139	8 43 58 141	10 41 57 94	21 45 87 135	12 77 97 136
159	1 3 7 33 10 53 72 92	14 42 98 144	5 27 71 95	11 59 96 119	16 54 89 134	13 49 107 125
160	1 3 7 23 5 51 101 141	13 39 91 139	8 41 79 107	9 27 63 74	12 29 43 87	10 25 93 128
166	1 5 39 140 17 79 109 142	3 15 88 117 6 42 112 152	7 35 107 150	11 32 116 129	9 19 77 95	2 46 113 121
168	1 3 16 125 6 18 42 72	4 38 117 149 7 21 77 140	11 37 104 129	10 41 81 110	5 52 85 146	8 53 73 159
169	1 3 17 126 11 24 122 143	6 18 80 102 4 69 79 118	7 32 41 70	5 92 115 134	8 76 91 121	20 53 81 117
170	1 3 16 116 18 44 68 151	7 21 112 132 4 32 131 140	5 27 80 164	12 47 72 141	10 61 92 128	8 74 97 114
171	1 3 9 60 20 44 99 130	5 15 45 134 18 53 124 143	16 48 121 144	12 80 113 149	13 39 117 146	7 21 84 95
172	1 3 10 130 22 56 82 119	11 31 114 137 4 96 108 144	6 57 151 159	15 33 62 100	5 59 84 133	14 30 91 131
173	1 3 10 77 18 65 100 122	4 12 40 135 25 79 110 143	5 32 144 157	19 60 105 149	6 23 62 121	11 37 83 103
174	1 3 7 127 8 27 70 99	5 15 35 113 14 53 71 93	13 36 82 150	11 56 84 133	12 44 86 153	9 34 119 157
175	1 3 7 94 11 33 75 140	9 27 63 146 5 85 128 152	8 61 77 134	10 96 113 127	15 34 55 131	12 25 51 155
176	1 3 8 34 12 28 137 156	9 27 72 130 10 54 75 161	14 94 129 153	11 49 71 147	6 56 97 140	4 93 106 123
177	1 3 9 83 7 21 54 65	5 15 45 61 23 85 110 134	4 26 104 164	12 50 126 157	18 37 106 142	27 68 120 149
178	1 3 7 133 10 72 107 157	9 27 63 129 5 16 103 150	14 70 92 153	12 38 148 161	8 90 113 154	15 34 74 94
179	1 3 7 22 24 67 104 129	10 30 41 70 12 57 108 143	13 39 91 107	17 49 76 141	14 42 98 132	5 23 69 102
180	1 3 7 146 9 68 106 162	11 33 77 166 8 32 107 128	19 42 88 137	13 39 67 117	12 70 127 163	15 55 116 164

Table A2. Optimal (v,6,1) CPCW codes.

Length	Codewords			
106	1 3 7 15 36	5 45 56 65 84	10 26 44 57 74	
107	1 3 7 12 89	8 34 47 63 83	10 27 41 69 84	
108	1 3 7 12 37	8 29 51 69 93	10 45 58 77 91	
109	1 3 7 12 39	8 28 41 58 83	10 24 53 72 88	
110	1 3 7 12 33	8 36 54 76 95	10 27 47 71 85	
111	1 3 7 12 29	8 24 56 66 96	13 33 47 74 93	
112	1 3 7 12 28	8 31 57 77 90	10 47 64 83 98	
113	1 3 7 12 25	8 27 43 60 81	10 36 51 65 93	
114	1 3 7 12 25	10 30 44 71 86	8 31 48 67 93	
115	1 3 7 12 25	8 37 54 81 95	10 26 45 77 92	
116	1 3 7 12 26	8 43 60 70 100	13 28 49 78 96	
117	1 3 7 12 22	13 33 50 74 92	8 31 47 77 91	
118	1 3 7 12 22	13 39 57 77 91	8 24 53 70 95	
119	1 3 7 12 25	8 28 43 62 89	10 42 59 82 98	
120	1 3 7 12 22	8 24 51 74 92	13 33 47 78 95	
125	1 3 19 31 54	5 47 64 93 104	7 41 50 77 110	4 67 80 105 119
127	1 3 8 51 113	6 38 61 96 108	4 20 33 60 101	10 34 52 88 116
128	1 3 10 30 53	6 21 45 87 100	4 18 35 40 72	8 33 59 71 117
129	1 3 9 25 97	5 15 60 80 98	12 52 73 102 115	4 23 71 82 99
130	1 3 7 15 107	9 41 57 75 119	5 33 50 72 93	10 35 71 84 111
131	1 3 23 36 105	6 65 73 97 116	9 37 54 79 93	4 11 16 57 87
132	1 3 10 26 87	5 57 74 95 108	11 33 73 100 112	6 14 50 78 97
133	1 3 7 25 95	5 48 78 105 125	9 26 49 68 80	11 46 61 75 112
134	1 3 7 92 107	8 48 66 101 113	9 46 63 82 120	5 25 84 95 108
135	1 3 7 38 61	8 24 34 56 107	5 17 30 50 93	9 27 66 80 95
136	1 3 7 31 47	8 49 74 88 124	10 52 65 103 125	5 64 83 109 118
137	1 3 7 20 56	8 46 58 83 122	9 44 72 104 115	10 26 78 96 123
138	1 3 7 15 48	9 43 64 96 116	5 24 62 80 108	10 23 49 60 121
139	1 3 7 16 57	11 42 60 74 111	10 48 68 103 115	5 22 66 99 118
140	1 3 7 15 59	11 35 48 86 112	5 30 46 79 113	9 62 80 109 130
141	1 3 7 15 102	9 19 35 53 128	5 33 60 89 110	11 41 58 78 103
142	1 3 7 15 46	9 29 61 87 114	10 35 75 86 108	5 23 93 106 123
143	1 3 7 15 47	5 27 43 62 112	9 29 77 101 118	10 55 83 94 120
144	1 3 7 15 93	5 39 81 118 128	17 46 73 91 121	9 22 50 109 133
145	1 3 7 12 33	8 57 74 111 125	13 44 73 97 120	16 59 81 99 126
146	1 3 7 15 105	16 34 54 86 119	5 40 62 87 115	10 29 55 68 79
147	1 3 7 15 44	11 39 56 69 90	5 24 40 72 105	9 31 83 93 129
148	1 3 7 12 42	10 56 88 115 135	15 34 51 72 96	18 44 73 98 126
149	1 3 7 12 102	8 34 71 122 139	13 32 55 75 96	14 30 45 103 127
150	1 3 7 12 45	8 39 87 124 134	13 54 69 86 136	19 49 72 107 129

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